I- Graphing Technology

Summary

Name	*General form*	*Function notation*
Linear	$f(x) = ax + b, \quad a \neq 0$	$f : x \mapsto ax + b, \quad a \neq 0$
Quadratic	$f(x) = ax^2 + bx + c, \quad a \neq 0$	$f : x \mapsto ax^2 + bx + c, \quad a \neq 0$
Cubic	$f(x) = ax^3 + bx^2 + cx + d, \quad a \neq 0$	$f : x \mapsto ax^3 + bx^2 + cx + d, \quad a \neq 0$
Exponential	$f(x) = a^x, \quad a > 0, \quad a \neq 1$	$f : x \mapsto a^x, \quad a > 0, \quad a \neq 1$
Logarithmic	$f(x) = \log_e x$ or $f(x) = \ln x$	$f : x \mapsto \log_e x$ or $f : x \mapsto \ln x$
Reciprocal	$f(x) = \frac{k}{x}, \quad x \neq 0, \quad k \neq 0$	$f : x \mapsto \frac{k}{x}, \quad x \neq 0, \quad k \neq 0$

Remark. *If we are asked to sketch a function, it will show the graph's general shape and its key features.*
If we are asked to draw a function, it should be done more carefully to scale.

Example

Consider the function $y = \frac{2^x}{x} - 2$. Use technology to help answer the following :

1. Find the axes intercepts.
2. Find any turning points of the function.
3. Find any asymptotes of the function.
4. State the domain and range of the function.
5. Sketch the function, showing its key features.

Example

1. Draw the functions $y = \dfrac{1}{x}$ and $y = 2 - x$ on the same set of axes, with the domain $-4 \leqslant x \leqslant 4$.
2. Hence find any solutions to the equation $2 - x = \dfrac{1}{x}$ on the domain $-4 \leqslant x \leqslant 4$.
3. On the domain $x \in \mathbb{R}$, for what values of k does $x + \dfrac{1}{x} = k$ have :
   (a) one solution
   (b) no solutions
   (c) two solutions ?

EXERCISE 1

For each of the functions given, use technology to answer the following:

i Find the axes intercepts.
ii Find any turning points of the function.
iii Find any asymptotes of the function.
iv State the domain and range of the function.
v Sketch the function, showing its key features.

a $y = \frac{1}{2}x(x-4)(x+3)$

b $y = \frac{4}{5}x^4 + 5x^3 + 5x^2 + 2$ on $-5 \leqslant x \leqslant 1$

c $y = x\,2^{-x}$

d $y = \dfrac{1-x^2}{(x+2)^2}$ on $-5 \leqslant x \leqslant 5$

e $y = \dfrac{1}{1+2^{-x}}$

f $y = \dfrac{x^2}{2^x} - 1$

EXERCISE 2

a Sketch the graphs of $f(x) = x^2 - \dfrac{4}{x}$ and $g(x) = -x^2 + 11$ for $-6 \leqslant x \leqslant 6$.

b One of the solutions to $f(x) = g(x)$ is $x \approx 2.51$. Use technology to determine the two negative solutions.

EXERCISE 3

Find the maximum value of:

a $y = -x^4 + 2x^3 + 5x^2 + x + 2$ on the interval $0 \leqslant x \leqslant 4$

b $y = -2x^4 + 5x^2 + x + 2$ on the interval:

i $-2 \leqslant x \leqslant 2$ ii $-2 \leqslant x \leqslant 0$ iii $0 \leqslant x \leqslant 2$.

EXERCISE 4

Sketch the graphs of $f(x) = e^{-x^2}$ and $g(x) = x^2$ on the same set of axes.
Hence solve $x^2 = e^{-x^2}$.

II- Transformation

a) Introduction

In particular, we can perform transformations of graphs to give the graph of a related function. These transformations include translations, stretches, and reflections.

We will consider transformations of the function $y = f(x)$ into :

- $y = f(x) + b$, b is a constant
- $y = pf(x)$, p is a positive constant
- $y = -f(x)$
- $y = f(x - a)$, a is a constant
- $y = f(qx)$, q is a positive constant
- $y = f(-x)$

Example

If $f(x) = x^2$, find in simplest form :

1. $f(2x)$
2. $f\left(\frac{x}{3}\right)$
3. $2f(x) + 1$
4. $f(x + 3) - 4$

EXERCISE 5

If $f(x) = x$, find in simplest form:

a $f(2x)$ b $f(x) + 2$ c $\frac{1}{2}f(x)$ d $2f(x) + 3$

EXERCISE 6

If $f(x) = x^3$, find in simplest form:

a $f(4x)$ b $\frac{1}{2}f(2x)$ c $f(x + 1)$ d $2f(x + 1) - 3$

EXERCISE 7

If $f(x) = 2^x$, find in simplest form:

a $f(2x)$ b $f(-x) + 1$ c $f(x - 2) + 3$ d $2f(x) + 3$

EXERCISE 8

If $f(x) = \frac{1}{x}$, find in simplest form:

a $f(-x)$ b $f(\frac{1}{2}x)$ c $2f(x) + 3$ d $3f(x - 1) + 2$

b) Translations

Property

For $y = f(x) + b$, the effect of b is to translate the graph vertically through b units.

- If $b > 0$ it moves upwards.
- If $b < 0$ it moves downwards.

Property

For $y = f(x - a)$, the effect of a is to translate the graph horizontally through a units.

- If $a > 0$ it moves to the right.
- If $a < 0$ it moves to the left.

Property

For $y = f(x - a) + b$, the graph is translated horizontally a units and vertically b units. We say it is translated by the vector $\begin{pmatrix} a \\ b \end{pmatrix}$.

EXERCISE 9

For each of the following functions f, sketch on the same set of axes the graphs of $y = f(x)$, $y = f(x) + 1$, and $y = f(x) - 2$.

a $f(x) = 2^x$ b $f(x) = x^3$ c $f(x) = \frac{1}{x}$ d $f(x) = (x - 1)^2$

EXERCISE 10

a On the same set of axes, graph $f(x) = x^2$, $y = f(x - 3)$, and $y = f(x + 2)$.

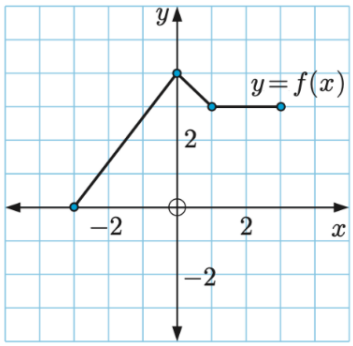
b What is the connection between the graphs of $y = f(x)$ and $y = f(x - a)$ if:

i $a > 0$ ii $a < 0$?

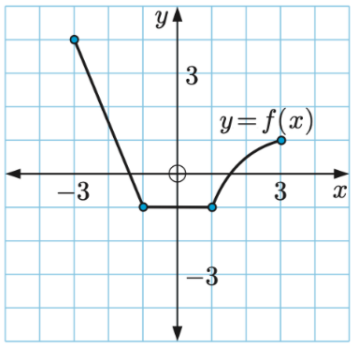
EXERCISE 11

Copy these functions and then draw the graph of $y = f(x - 2) - 3$.

a



b



c) Stretches

Investigation

1. For $f(x) = x + 2$, find in simplest form :
   - **i** $3f(x)$
   - **ii** $\frac{1}{2}f(x)$
   - **iii** $5f(x)$
2. Graph all four functions on the same set of axes.
3. What effect does the constant p have when $y = f(x)$ is transformed to $y = pf(x)$, $p > 0$?

1. For $f(x) = x^2$, find in simplest form :
   (a) $f(2x)$
   (b) $f(3x)$
   (c) $f\left(\frac{x}{4}\right)$
2. Graph all four functions on the same set of axes.
3. What effect does the constant q have when $y = f(x)$ is transformed to $y = f(qx)$, $q > 0$?

Property

For $y = pf(x)$, $p > 0$, the effect of p is to vertically stretch the graph by the scale factor p.

- If $p > 1$ it moves points of $y = f(x)$ further away from the x-axis.
- If $0 < p < 1$ it moves points of $y = f(x)$ closer to the x-axis.

Property

For $y = f(qx)$, $q > 0$, the effect of q is to horizontally stretch the graph by the scale factor $\frac{1}{q}$.

- If $q > 1$ it moves points of $y = f(x)$ closer to the y-axis.
- If $0 < q < 1$ it moves points of $y = f(x)$ further away from the y-axis.

EXERCISE 12

Sketch, on the same set of axes, the graphs of $y = f(x)$, $y = 2f(x)$, and $y = 3f(x)$ for each of:

a $f(x) = x^2$ b $f(x) = x^3$ c $f(x) = e^x$

d $f(x) = \ln x$ e $f(x) = \frac{1}{x}$

EXERCISE 13

Sketch, on the same set of axes, the graphs of $y = f(x)$ and $y = f(2x)$ for each of:

a $y = x^2$ b $y = (x-1)^2$ c $y = (x+3)^2$

EXERCISE 14

Sketch, on the same set of axes, the graphs of $y = f(x)$ and $y = f(\frac{x}{2})$ for each of:

a $y = x^2$ b $y = 2x$ c $y = (x+2)^2$

EXERCISE 15

Consider the function $f : x \mapsto x^2$.
On the same set of axes sketch the graphs of:

a $y = f(x)$, $y = 3f(x-2)+1$, and $y = 2f(x+1)-3$

b $y = f(x)$, $y = f(x-3)$, $y = f(\frac{x}{2}-3)$, $y = 2f(\frac{x}{2}-3)$, and $y = 2f(\frac{x}{2}-3)+4$

c $y = f(x)$ and $y = \frac{1}{4}f(2x+5)+1$.

d) Reflections

Investigation

1. Consider $f(x) = x^3 - 2$.
   (a) Find in simplest form :
      i. $-f(x)$
      ii. $f(-x)$
   (b) Graph $y = f(x)$, $y = -f(x)$, and $y = f(-x)$ on the same set of axes.
2. Consider $f(x) = e^x$.
   (a) Find in simplest form :
      i. $-f(x)$
      ii. $f(-x)$
   (b) Graph $y = f(x)$, $y = -f(x)$, and $y = f(-x)$ on the same set of axes.
3. What transformation moves :
   (a) $y = f(x)$ to $y = -f(x)$
   (b) $y = f(x)$ to $y = f(-x)$?

Property

For $y = -f(x)$, we reflect $y = f(x)$ in the x-axis.

Property

For $y = f(-x)$, we reflect $y = f(x)$ in the y-axis.

EXERCISE 16

On the same set of axes, sketch the graphs of $y = f(x)$ and $y = -f(x)$ for:

a $f(x) = 3x$ b $f(x) = e^x$ c $f(x) = x^2$

d $f(x) = \ln x$ e $f(x) = x^3 - 2$ f $f(x) = 2(x+1)^2$

EXERCISE 17

a Find $f(-x)$ for:

i $f(x) = 2x + 1$ ii $f(x) = x^2 + 2x + 1$ iii $f(x) = x^3$

b Graph $y = f(x)$ and $y = f(-x)$ for:

i $f(x) = 2x + 1$ ii $f(x) = x^2 + 2x + 1$ iii $f(x) = x^3$

EXERCISE 18

The function $f(x) = x^3 - \ln x$ is reflected in the x-axis to $g(x)$. Find $g(x)$.

EXERCISE 19

The function $f(x) = x^4 - 2x^3 - 3x^2 + 5x - 7$ is reflected in the y-axis to $g(x)$. Find $g(x)$.

EXERCISE 20

The function $y = f(x)$ is transformed to $h(x) = f(-x)$.

a Find the image points on $h(x)$ for the following points on $f(x)$:

i $(2, -1)$ ii $(0, 3)$ iii $(-1, 2)$

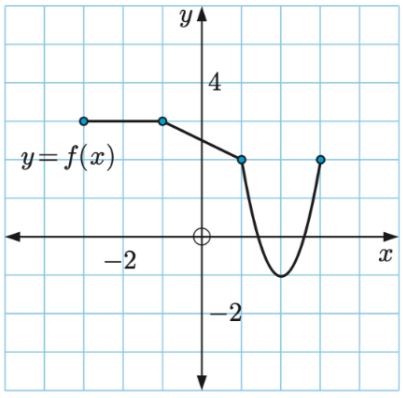
b Find the points on $f(x)$ corresponding to the following points on $h(x)$:

i $(5, -4)$ ii $(0, 3)$ iii $(2, 3)$

EXERCISE 21

Copy the graph of $y = f(x)$ alongside, then draw the graph of:

a $y = -f(x)$ b $y = f(-x)$



EXERCISE 22

A function $f(x)$ is transformed to the function $g(x) = -f(-x)$.

a Describe the nature of the transformation.

b If $(3, -7)$ lies on $y = f(x)$, find the transformed point on $y = g(x)$.

c Find the point on $f(x)$ that transforms to the point $(-5, -1)$.

III- Miscellaneaous transformations

Example

Consider $f(x) = \frac{1}{2}x + 1$. On separate sets of axes graph :

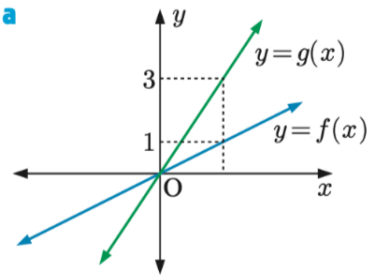
1. $y = f(x)$ and $y = f(x + 2)$
2. $y = f(x)$ and $y = f(x) + 2$
3. $y = f(x)$ and $y = 2f(x)$
4. $y = f(x)$ and $y = -f(x)$

EXERCISE 23

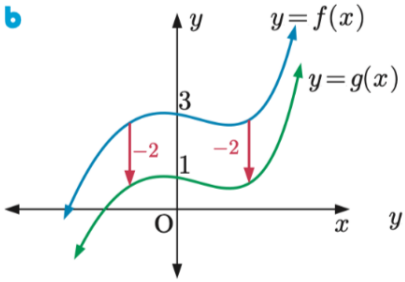
On each of the following $f(x)$ is mapped onto $g(x)$ using a single transformation.

i Describe the transformation fully. ii Write $g(x)$ in terms of $f(x)$.

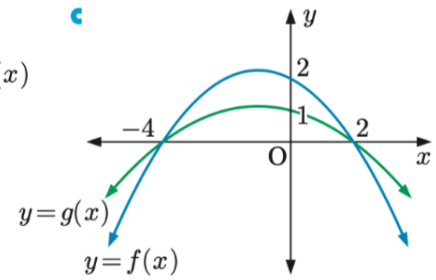
a



b



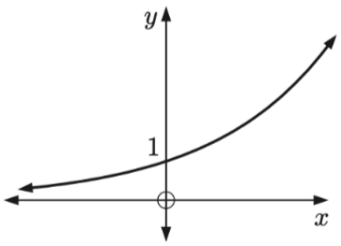
c



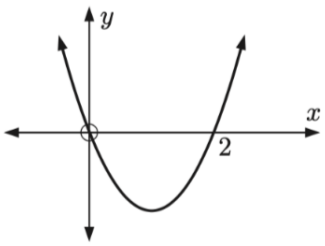
EXERCISE 24

Copy the following graphs for $y = f(x)$ and sketch the graphs of $y = -f(x)$ on the same axes.

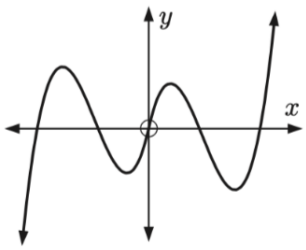
a



b



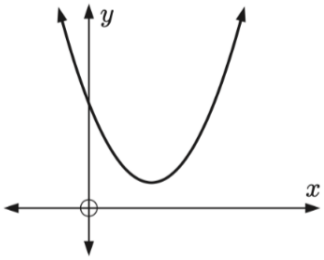
c



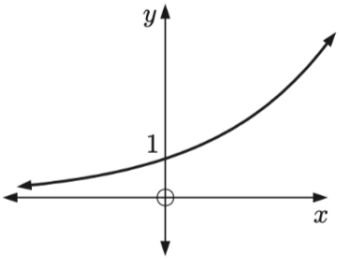
EXERCISE 25

Given the following graphs of $y = f(x)$, sketch graphs of $y = f(-x)$:

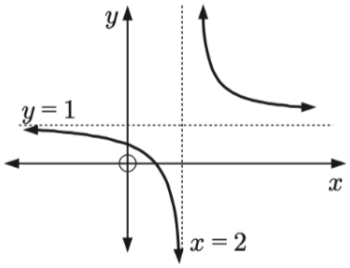
a



b



c

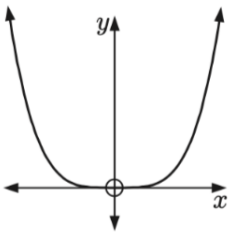


EXERCISE 26

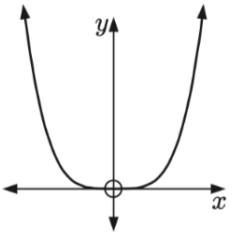
The scales on the graphs below are the same. Match each equation to its graph.

A $y = x^4$ **B** $y = 2x^4$ **C** $y = \frac{1}{2}x^4$ **D** $y = 6x^4$

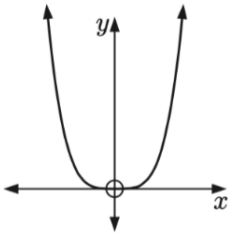
a



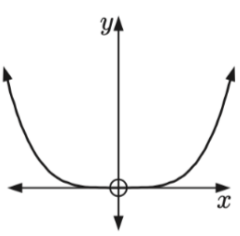
b



c



d



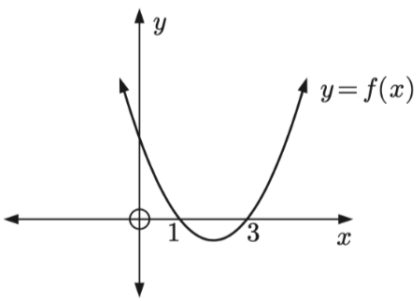
EXERCISE 27

For the graph of $y = f(x)$ given, sketch the graph of:

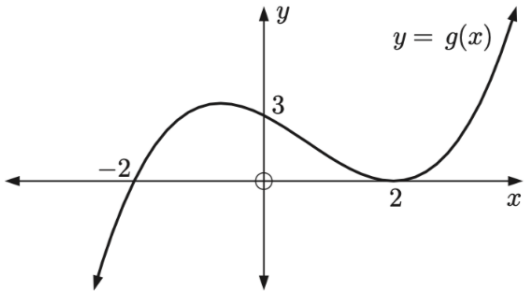
a $y = 2f(x)$ **b** $y = \frac{1}{2}f(x)$

c $y = f(x + 2)$ **d** $y = f(2x)$

e $y = f(\frac{1}{2}x)$



EXERCISE 28



For the graph of $y = g(x)$ given, sketch the graph of:

a $y = g(x) + 2$ **b** $y = -g(x)$

c $y = g(-x)$ **d** $y = g(x + 1)$

Example

3. Consider $f : x \mapsto \dfrac{2^x}{x}$.
   (a) Does the function have any axes intercepts ?
   (b) Find the equations of the asymptotes of the function.
   (c) Find any turning points of the function.
   (d) Sketch the function for $-4 \leqslant x \leqslant 4$.
4. Consider $f : x \mapsto 2^{-x}$.
   (a) Use your calculator to help graph the function.
   (b) True or false ?
      i. As $x \to \infty$, $2^{-x} \to 0$.
      ii. As $x \to -\infty$, $2^{-x} \to 0$.
      iii. The y-intercept is $\dfrac{1}{2}$.
      iv. $2^{-x} > 0$ for all x.